

NETWORKS AND COMPLEXITY

Solution 19-3

*This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 19.3: Linked pair [Lvl. 2]

Let's consider the linked pair (o-o) network in more detail.

- a) Write the adjacency matrix $\mathbf{A}$ and compute the eigenvalues λ_n , the right eigenvectors $\mathbf{v}_n$ and the left eigenvectors $\mathbf{w}_n^\dagger$.

Solution

The adjacency matrix is

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (1)$$

The trace and determinant of this matrix tell us that the eigenvalues sum to zero and multiply to -1, so they are

$$\lambda_1 = 1, \quad \lambda_2 = -1. \quad (2)$$

The right eigenvectors are

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \quad (3)$$

Since the matrix is symmetric, we have $\mathbf{v}_n = \mathbf{w}_n$, and hence

$$\mathbf{w}_1^\dagger = (1, 1), \quad \mathbf{w}_2^\dagger = (1, -1), \quad (4)$$

- b) Consider the iteration $\mathbf{v}(i+1) = \mathbf{A}\mathbf{v}(i)$, starting with $\mathbf{v}(0) = (1, 2)^\top$. Find a closed-form solution for $\mathbf{v}(i)$.

Solution

We remember from Chap. 12 how this is done. We start with the iteration rule

$$\mathbf{v}(i+1) = \mathbf{A}\mathbf{v}(i) \quad (5)$$

and formally decompose the vector $\mathbf{v}(i)$ into right eigenvectors of $\mathbf{A}$,

$$\mathbf{v}(i) = \sum c_n(i) \mathbf{v}_n, \quad (6)$$

substituting into the iteration rule yields

$$\sum c_n(i+1) \mathbf{v}_n = \mathbf{A} \sum c_n(i) \mathbf{v}_n = \sum c_n(i) \lambda_n \mathbf{v}_n. \quad (7)$$

Since the two eigenvectors point in different directions, we can consider the two components in the directions of the two eigenvectors independently, which yields

$$c_1(i+1) = c_1(i) \quad (8)$$

$$c_2(i+1) = -c_2(i) \quad (9)$$

So in each step we leave c_1 as it is and multiply c_2 with -1. Hence

$$c_1(i) = c_1(0) \quad c_2(i) = (-1)^i c_2(0) \quad (10)$$

Now we only need to determine $c_1(0)$ and $c_2(0)$ by decomposing the initial $\mathbf{v}(0) = (1, 2)^T$. We can do this with our shiny new trick of multiplying the initial vector with the left eigenvectors. Note however that we haven't normalized our eigenvectors properly, so we additionally need to divide by $\mathbf{w}_n^\dagger \mathbf{v}_n$ to compensate

$$c_1 = \frac{\mathbf{w}_1^\dagger \mathbf{v}(0)}{\mathbf{w}_1^\dagger \mathbf{v}_1} = \frac{(1, 1) \begin{pmatrix} 1 \\ 2 \end{pmatrix}}{(1, 1) \begin{pmatrix} 1 \\ 1 \end{pmatrix}} = \frac{3}{2} \quad (11)$$

$$c_2 = \frac{\mathbf{w}_2^\dagger \mathbf{v}(0)}{\mathbf{w}_2^\dagger \mathbf{v}_2} = \frac{(1, -1) \begin{pmatrix} 1 \\ 2 \end{pmatrix}}{(1, -1) \begin{pmatrix} 1 \\ -1 \end{pmatrix}} = -\frac{1}{2} \quad (12)$$

Let's check if this is right

$$\mathbf{v}(0) = \frac{3}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad (13)$$

which is what we wanted. Putting everything together we have the solution

$$\mathbf{v}(i) = \frac{3}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \frac{(-1)^i}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (14)$$

So if i is even

$$\mathbf{v}(i) = \frac{3}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad (15)$$

and if i is odd

$$\mathbf{v}(i) = \frac{3}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}. \quad (16)$$

Hence, our vector $\mathbf{v}$ will alternate between these values for all eternity.

- c) Now consider the shifted matrix $\mathbf{B} = \mathbf{A} + \mathbf{I}$, where $\mathbf{I}$ is an identity matrix of appropriate size. Again, find the eigenvalues and eigenvectors and study the iteration $\mathbf{v}(i+1) = \mathbf{B}\mathbf{v}(i)$, starting from $\mathbf{v}(0) = (1, 2)^T$.

Solution

The identity matrix is

$$\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (17)$$

Hence,

$$\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad (18)$$

Since this is a shift by one unit we expect the eigenvalues to be $\lambda_1 = 2$ and $\lambda_2 = 0$. The shift leaves the eigenvectors unchanged, so we still have

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \mathbf{w}_1^\dagger = (1, 1) \quad (19)$$

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \mathbf{w}_2^\dagger = (1, -1) \quad (20)$$

Since the eigenvectors haven't changed, the decomposition of the initial state hasn't changed as well so

$$c_1(0) = \frac{3}{2} \quad c_2(0) = -\frac{1}{2} \quad (21)$$

What has changed, however, is the time evolution of these coefficients. If we take the iteration rule

$$\mathbf{v}(i+1) = \mathbf{B}\mathbf{v}(i) \quad (22)$$

and substitute the eigenvector decomposition

$$\mathbf{v}(i) = \sum c_n(i) \mathbf{v}_n \quad (23)$$

we have as before

$$\sum c_n(i+1) \mathbf{v}_n = \mathbf{B} \sum c_n(i) \mathbf{v}_n = \sum c_n(i) \lambda_n \mathbf{v}_n \quad (24)$$

but if we split this into two equations for the individual c_n and substitute the eigenvectors in we now have

$$c_1(i+1) = 2c_1(i) \quad (25)$$

$$c_2(i+1) = 0 \quad (26)$$

Hence, we find the solution

$$\mathbf{v}(i) = \frac{3}{2} 2^i \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (27)$$

for $i > 0$. So already after one iteration we have ($i = 1$)

$$\mathbf{v}(1) = \begin{pmatrix} 3 \\ 3 \end{pmatrix} \quad (28)$$

which is an eigenvector of $\mathbf{A}$. So already after one step the iteration converges to the desired eigenvector. This happens so quickly in this case because the 'error', i.e., the component in the direction of the second eigenvector is multiplied by 0 in every step, with this particular shift.